

| SEQ NO | REV | NOTE NO. | ITEM NAME   | ITEM CODE | SIZE RANGE / SMALL END SIZES |     | DESCRIPTION                                                                                                      | REFERENCE / LARGE END SIZES |   | NOTES |                                                                                                                                          |
|--------|-----|----------|-------------|-----------|------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----------------------------|---|-------|------------------------------------------------------------------------------------------------------------------------------------------|
| 10     |     |          |             |           |                              |     | *****VALVES*****                                                                                                 |                             |   | 810   | NIPPLE TO BE CUT FROM SCH 160 PIPE.                                                                                                      |
| 20     |     |          |             |           |                              |     |                                                                                                                  |                             |   | 817   | USE ONLY FOR TIE IN                                                                                                                      |
| 20     |     |          |             |           |                              |     | ***** MODULAR VALVES *****                                                                                       |                             |   | 821   | TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.                                                                         |
| 30     |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 30     | U04 | VAW      | MODULAR DBB | M03EC02   | 0.75                         | 1   | MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR | 1.5                         | 2 | 823   | USE OF THIS ITEM IS LIMITED TO MAXIMUM DESIGN TEMPERATURE OF 250°C                                                                       |
| 40     |     |          |             |           |                              |     |                                                                                                                  |                             |   | B1    | ALL BOLTS ARE METRIC DIA.                                                                                                                |
| 40     |     | VB2      |             |           |                              |     | ***** BALL VALVES *****                                                                                          |                             |   | B2    | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10                                                  |
| 50     |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 50     | U04 |          | BALL        | A01AC01   | 3                            | 4   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO                                                   | 0                           | 0 |       |                                                                                                                                          |
| 60     | U04 |          | BALL        | A01AC02   | 3                            | 4   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO                                                  | 0                           | 0 | B6    | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22                                                  |
| 70     | U04 |          | BALL        | A01AC03   | 6                            | 6   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO                                                   | 0                           | 0 |       |                                                                                                                                          |
| 80     | U04 |          | BALL        | A01AC04   | 6                            | 6   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB GO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 90     | U04 |          | BALL        | A01AC07   | 8                            | 8   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO                                                   | 0                           | 0 | FA8   | USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.                                                                                    |
| 100    | U04 |          | BALL        | A01AC08   | 8                            | 8   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 110    | U04 |          | BALL        | B01AC01   | 0.5                          | 3   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                                   | 0                           | 0 | FF5   | FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES                                                                           |
| 120    | U04 |          | BALL        | B01AC02   | 0.5                          | 3   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 130    | U04 | VA5      | BALL        | B01AC03   | 4                            | 4   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO                                                   | 0                           | 0 | FS1   | USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.                                                          |
| 140    | U04 | VA5      | BALL        | B01AC03   | 4                            | 4   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                                   | 0                           | 0 |       |                                                                                                                                          |
| 150    | U04 | VA5      | BALL        | B01AC04   | 4                            | 4   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 160    | U04 | VA5      | BALL        | B01AC04   | 6                            | 6   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                                  | 0                           | 0 | G1    | NOT TO BE USED ON FGP                                                                                                                    |
| 170    | U04 |          | BALL        | B01AC07   | 6                            | 24  | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO                                                   | 0                           | 0 | GA6   | SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117                                                                                  |
| 180    | U04 |          | BALL        | B01AC08   | 6                            | 24  | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 190    | U04 | VA5      | BALL        | B01JC07   | 26                           | 48  | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A                                   | 0                           | 0 | ND9   | USE ONLY WITH CONSENT OF MATERIALS ENGINEER.                                                                                             |
| 200    | U04 | VA5      | BALL        | B01JC08   | 26                           | 48  | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A                                  | 0                           | 0 |       |                                                                                                                                          |
| 210    | U04 | VA5      | BALL        | B01AC96   | 0.75                         | 2   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO 2800MM EXT                                        | 0                           | 0 | SB6   | USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.                                                      |
| 220    | U04 | VAQ      | BALL        | BA1C1     | 1                            | 1   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FS RB LONG PATT                                                       | 0                           | 0 |       |                                                                                                                                          |
| 230    | U04 |          | BALL        | B03AC01   | 0.5                          | 3   | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                                   | 0                           | 0 |       |                                                                                                                                          |
| 240    | U04 | VA5      | BALL        | B03AC02   | 0.5                          | 2   | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                                  | 0                           | 0 | UG1   | PTFE GASKET ONLY TO BE USED FOR CIVIL WSC SYSTEMS AND INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING.                             |
| 250    | U04 | VA5      | BALL        | A03AC01   | 3                            | 3   | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO                                                   | 0                           | 0 |       |                                                                                                                                          |
| 260    | U04 | VA5      | BALL        | A03AC02   | 3                            | 3   | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO                                                  | 0                           | 0 |       |                                                                                                                                          |
| 270    |     |          |             |           |                              |     | *****BUTTERFLY VALVE*****                                                                                        |                             |   | UG2   | BOLTS ONLY TO BE USED FOR CIVIL WSC SYSTEMS AND INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING.                                   |
| 270    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 280    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 280    |     |          | BUTTERFLY   | T01AC61   | 10                           | 24  | BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO                                        | 0                           | 0 |       |                                                                                                                                          |
| 290    |     |          | BUTTERFLY   | T01JC61   | 26                           | 48  | BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A                   | 0                           | 0 | UJ2   | USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.                        |
| 300    |     |          |             |           |                              |     | ***** GATE VALVES *****                                                                                          |                             |   |       |                                                                                                                                          |
| 300    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 310    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 310    | U04 |          | GATE        | G01AC48   | 0.5                          | 1.5 | GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW                                                                 | 0                           | 0 | UJ3   | USE 3IN WITH THERMOWELL CONNECTIONS ONLY                                                                                                 |
| 320    | U04 | VA5      | GATE        | G01AC41   | 10                           | 16  | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                                      | 0                           | 0 |       |                                                                                                                                          |
| 330    | U04 |          | GATE        | G01AC41   | 2                            | 8   | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                                      | 0                           | 0 | UJ4   | FOR 2 INCH THERMOWELL CONNECTIONS ONLY                                                                                                   |
| 340    | U04 | VA5      | GATE        | G01AC42   | 18                           | 24  | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO                                                                      | 0                           | 0 |       |                                                                                                                                          |
| 350    | U04 |          | GATE        | G03AC48   | 0.5                          | 1.5 | GATE 300# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW                                                                 | 0                           | 0 | UK9   | DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.                                                                                            |
| 360    | U04 |          | GATE        | G03AC41   | 2                            | 6   | GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                                      | 0                           | 0 |       |                                                                                                                                          |
| 370    |     |          |             |           |                              |     | ***** THROUGH CONDUIT GATE VALVES *****                                                                          |                             |   | VA5   | USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS                                                                  |
| 370    |     |          |             |           |                              |     |                                                                                                                  |                             |   | VAQ   | FOR TURN AROUND TIE-INS ONLY                                                                                                             |
| 380    | U04 | VA5      | GATE        | W01AC70   | 12                           | 16  | GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW                                                | 0                           | 0 | VAW   | VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.                              |
| 390    | U04 | VA5      | GATE        | W01AC71   | 18                           | 24  | GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO                                                | 0                           | 0 |       |                                                                                                                                          |
| 400    |     |          |             |           |                              |     | ***** GLOBE VALVES *****                                                                                         |                             |   | VB2   | SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.                                                                     |
| 400    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 410    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 410    | U04 |          | GLOBE       | L01AC48   | 0.75                         | 1.5 | GLOBE 150# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW                                                                 | 0                           | 0 | VC1   | SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. |
| 420    |     |          | GLOBE       | L01AC41   | 2                            | 6   | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW                                                                      | 0                           | 0 |       |                                                                                                                                          |
| 430    |     |          | GLOBE       | L01AC42   | 8                            | 10  | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO                                                                      | 0                           | 0 |       |                                                                                                                                          |
| 440    |     |          |             |           |                              |     | ***** CHECK VALVES *****                                                                                         |                             |   |       |                                                                                                                                          |
| 440    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 450    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 450    | U04 | VC1      | CHECK       | C01AC50   | 0.5                          | 1.5 | CHECK PISTON 150# RF LTCS A350 LF2 CL.1 TRIM 16                                                                  | 0                           | 0 |       |                                                                                                                                          |
| 460    | U04 |          | CHECK       | C01AC45   | 2                            | 6   | CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED                                                  | 0                           | 0 |       |                                                                                                                                          |
| 470    | U04 |          | CHECK       | C01AC52   | 8                            | 24  | CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16                                                                 | 0                           | 0 |       |                                                                                                                                          |
| 480    | U04 |          | CHECK       | C01JC52   | 26                           | 48  | CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A                                                 | 0                           | 0 |       |                                                                                                                                          |
| 490    |     |          |             |           |                              |     | *****PIPES*****                                                                                                  |                             |   |       |                                                                                                                                          |
| 490    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 500    |     |          |             |           |                              |     |                                                                                                                  |                             |   |       |                                                                                                                                          |
| 500    | U04 |          | PIPE        | 5358508   | 0.5                          | 1.5 | PIPE SCH XS SMLS LTCS A333 GR.6 PE                                                                               | 0                           | 0 |       |                                                                                                                                          |
| 510    |     |          | PIPE        | 5358514   | 2                            | 2   | PIPE SCH XS SMLS LTCS A333 GR.6 BE                                                                               | 0                           | 0 |       |                                                                                                                                          |
| 520    |     |          | PIPE        | 5358505   | 3                            | 14  | PIPE SCH STD SMLS LTCS A333 GR.6 BE                                                                              | 0                           | 0 |       |                                                                                                                                          |
| 530    |     |          | PIPE        | 5358501   | 16                           | 16  | PIPE SCH 20 SMLS LTCS A333 GR.6 BE                                                                               | 0                           | 0 |       |                                                                                                                                          |
| 540    |     |          | PIPE        | 5358597   | 18                           | 24  | PIPE SCH 20 EFW LTCS A671 C060 CL.22 BE                                                                          | 0                           | 0 |       |                                                                                                                                          |
| 550    |     |          | PIPE        | 5358576   | 26                           | 38  | PIPE SCH STD EFW LTCS A671 C060 CL.22 BE                                                                         | 0                           | 0 |       |                                                                                                                                          |

|                                                                                    |                                                |                                                                                                                                                  |                     |                                                                                                                         |                                                                     |            |            |     |                |                   |
|------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROIL<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                     | CORR.ALLOW.         | SERVICE                                                                                                                 | MATERIAL                                                            | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                                | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 1<br>of 4 | 150H21            |

| SEQ NO | REV | NOTE NO. | ITEM NAME | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                   | REFERENCE / LARGE END SIZES |      | NOTES |  |
|--------|-----|----------|-----------|-----------|------------------------------|------|-----------------------------------------------|-----------------------------|------|-------|--|
| 560    |     |          | PIPE      | 5358577   | 40                           | 48   | PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE       | 0                           | 0    |       |  |
| 570    |     | 810      | PIPE      | 5358512   | 0.5                          | 0.5  | PIPE SCH 160 SMLS LTCS A333 GR.6 PE           | 0                           | 0    |       |  |
| 580    | U04 | UK9      | PIPE      | 5358511   | 0.5                          | 0.75 | PIPE SCH XXS SMLS LTCS A333 GR.6 PE           | 0                           | 0    |       |  |
| 590    | U04 | 821      | PIPE      | 5358747   | 24                           | 24   | PIPE WT 19.05MM EFW LTCS A671 CC60 CL.22 BE   | 0                           | 0    |       |  |
| 600    |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 600    |     |          |           |           |                              |      | ***** FITTINGS *****                          |                             |      |       |  |
| 610    |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 610    |     |          | ELBOW     | 5392600   | 0.5                          | 2    | ELL BW 90 DEG LR SCH XS LTCS A420 WPL6        | 0                           | 0    |       |  |
| 620    | U04 | UK9      | ELBOW     | 5392815   | 0.5                          | 0.75 | ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6       | 0                           | 0    |       |  |
| 630    |     |          | ELBOW     | 5392438   | 3                            | 14   | ELL BW 90 DEG LR SCH STD LTCS A420 WPL6       | 0                           | 0    |       |  |
| 640    |     |          | ELBOW     | 5392276   | 16                           | 16   | ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6        | 0                           | 0    |       |  |
| 650    |     |          | ELBOW     | 5392280   | 18                           | 24   | ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W      | 0                           | 0    |       |  |
| 660    |     |          | ELBOW     | 5392241   | 24                           | 24   | ELL BW 90 DEG SR SCH 20 LTCS A420 WPL6-W      | 0                           | 0    |       |  |
| 670    |     |          | ELBOW     | 5392496   | 26                           | 38   | ELL 90 DEG LR SCH STD LTCS A420 WPL6-W        | 0                           | 0    |       |  |
| 680    |     |          | ELBOW     | 5392497   | 40                           | 48   | ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W      | 0                           | 0    |       |  |
| 690    | U04 | 821      | ELBOW     | 5392489   | 24                           | 24   | ELL BW 90 DEG LR WT 19.05 MM LTCS A420 WPL6-W | 0                           | 0    |       |  |
| 700    | U04 |          | ELBOW     | 5392606   | 0.5                          | 2    | ELL BW 45 DEG LR SCH XS LTCS A420 WPL6        | 0                           | 0    |       |  |
| 710    | U04 | UK9      | ELBOW     | 5392831   | 0.5                          | 0.75 | ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6       | 0                           | 0    |       |  |
| 720    |     |          | ELBOW     | 5392444   | 3                            | 14   | ELL BW 45 DEG LR SCH STD LTCS A420 WPL6       | 0                           | 0    |       |  |
| 730    |     |          | ELBOW     | 5392282   | 16                           | 16   | ELL BW 45 DEG SCH 20 LTCS A420 WPL6           | 0                           | 0    |       |  |
| 740    |     |          | ELBOW     | 5392281   | 18                           | 24   | ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W      | 0                           | 0    |       |  |
| 750    |     |          | ELBOW     | 5392607   | 26                           | 38   | ELL BW 45 DEG SCH STD LTCS A420 WPL6-W        | 0                           | 0    |       |  |
| 760    |     |          | ELBOW     | 5392608   | 40                           | 48   | ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W      | 0                           | 0    |       |  |
| 770    |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 770    | U04 |          | TEE       | 5392609   | 0.5                          | 2    | TEE BW SCH XS LTCS A420 WPL6                  | 0                           | 0    |       |  |
| 780    |     |          | TEE       | 5392447   | 3                            | 14   | TEE BW SCH STD LTCS A420 WPL6                 | 0                           | 0    |       |  |
| 790    |     |          | TEE       | 5392285   | 16                           | 16   | TEE BW SCH 20 LTCS A420 WPL6                  | 0                           | 0    |       |  |
| 800    |     |          | TEE       | 5392295   | 18                           | 24   | TEE BW SCH 20 LTCS A420 WPL6-W                | 0                           | 0    |       |  |
| 810    |     |          | TEE       | 5392405   | 26                           | 38   | TEE SCH STD LTCS A420 WPL6-W                  | 0                           | 0    |       |  |
| 820    |     |          | TEE       | 5392406   | 40                           | 48   | TEE BW SCH XS LTCS A420 WPL6-W                | 0                           | 0    |       |  |
| 830    | U04 | 821      | TEE       | 5392253   | 24                           | 24   | TEE BW WT 19.05MM LTCS A420 WPL6-W            | 0                           | 0    |       |  |
| 840    | U04 | UK9      | TEE       | 5392879   | 0.5                          | 0.75 | TEE SCH XXS LTCS A420 WPL6                    | 0                           | 0    |       |  |
| 850    | U04 |          | TEE RED   | 5392661   | 0.5                          | 1.5  | TEE RED BW SCH XS LTCS A420 WPL6              | 1                           | 2    |       |  |
| 860    |     |          | FLANGE    | 5392454   | 1.5                          | 2    | TEE RED BW SCH STD/XS LTCS A420 WPL6          | 3                           | 4    |       |  |
| 870    |     |          | TEE RED   | 5392445   | 3                            | 12   | TEE RED BW SCH STD LTCS A420 WPL6             | 4                           | 14   |       |  |
| 880    |     |          | TEE RED   | 5392289   | 6                            | 14   | TEE RED BW SCH 20/STD LTCS A420 WPL6          | 16                          | 16   |       |  |
| 890    |     |          | TEE RED   | 5392296   | 8                            | 14   | TEE RED BW SCH 20/STD LTCS A420 WPL6-W        | 18                          | 24   |       |  |
| 900    |     |          | TEE RED   | 5392428   | 14                           | 14   | TEE RED BW SCH STD LTCS A420 WPL6-W           | 26                          | 32   |       |  |
| 910    |     |          | TEE RED   | 5392428   | 26                           | 36   | TEE RED BW SCH STD LTCS A420 WPL6-W           | 28                          | 38   |       |  |
| 920    |     |          | TEE RED   | 5392292   | 16                           | 20   | TEE RED BW SCH 20 LTCS A420 WPL6-W            | 18                          | 24   |       |  |
| 930    |     |          | TEE RED   | 5392320   | 16                           | 24   | TEE RED SCH STD/20 LTCS A420 WPL6-W           | 26                          | 38   |       |  |
| 940    |     |          | TEE RED   | 5392475   | 18                           | 24   | TEE RED BW SCH XS/20 LTCS A420 WPL6-W         | 40                          | 40   |       |  |
| 950    |     |          | TEE RED   | 5392476   | 36                           | 38   | TEE RED SCH XS/STD LTCS A420 WPL6-W           | 40                          | 42   |       |  |
| 960    |     |          | TEE RED   | 5392413   | 40                           | 46   | TEE RED SCH XS LTCS A420 WPL6-W               | 44                          | 48   |       |  |
| 970    |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 970    | U04 |          | CAP       | 5392610   | 0.5                          | 2    | CAP BW SCH XS LTCS A420 WPL6                  | 0                           | 0    |       |  |
| 980    |     |          | CAP       | 5392610   | 40                           | 48   | CAP BW SCH XS LTCS A420 WPL6                  | 0                           | 0    |       |  |
| 990    |     |          | CAP       | 5392448   | 3                            | 14   | CAP BW SCH STD LTCS A420 WPL6                 | 0                           | 0    |       |  |
| 1000   |     |          | CAP       | 5392448   | 26                           | 38   | CAP BW SCH STD LTCS A420 WPL6                 | 0                           | 0    |       |  |
| 1010   |     |          | CAP       | 5392286   | 16                           | 24   | CAP BW SCH 20 LTCS A420 WPL6                  | 0                           | 0    |       |  |
| 1020   |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 1020   |     |          | RED CONC  | 5392458   | 0.5                          | 1.5  | REDUCER CONC SCH XS LTCS A420 WPL6            | 0.75                        | 2    |       |  |
| 1030   | U04 |          | RED CONC  | 5392452   | 1.5                          | 2.5  | REDUCER CONC SCH STD/XS LTCS A420 WPL6        | 3                           | 4    |       |  |
| 1040   |     |          | RED CONC  | 5392449   | 3                            | 12   | REDUCER CONC SCH STD LTCS A420 WPL6           | 4                           | 14   |       |  |
| 1050   |     |          | RED CONC  | 5392290   | 8                            | 14   | REDUCER CONC SCH 20/STD LTCS A420 WPL6        | 16                          | 16   |       |  |
| 1060   |     |          | RED CONC  | 5392297   | 10                           | 14   | REDUCER CONC SCH 20/STD LTCS A420 WPL6-W      | 18                          | 20   |       |  |
| 1070   |     |          | RED CONC  | 5392287   | 16                           | 20   | REDUCER CONC SCH 20 LTCS A420 WPL6-W          | 18                          | 24   |       |  |
| 1080   |     |          | RED CONC  | 5392500   | 18                           | 24   | REDUCER CONC SCH STD/20 LTCS A420 WPL6-W      | 26                          | 36   |       |  |
| 1090   |     |          | RED CONC  | 5392505   | 26                           | 36   | REDUCER CONC SCH STD LTCS A420 WPL6-W         | 28                          | 38   |       |  |
| 1100   |     |          | RED CONC  | 5392506   | 30                           | 38   | REDUCER CONC SCH XS/STD LTCS A420 WPL6-W      | 40                          | 48   |       |  |
| 1110   |     |          | RED CONC  | 5392507   | 40                           | 46   | REDUCER CONC SCH XS LTCS A420 WPL6-W          | 42                          | 48   |       |  |
| 1120   |     |          | RED ECC   | 5392461   | 0.5                          | 1.5  | REDUCER ECC SCH XS LTCS A420 WPL6             | 0.75                        | 2    |       |  |
| 1130   |     |          | RED ECC   | 5392453   | 1.5                          | 2    | REDUCER ECC SCH STD/XS LTCS A420 WPL6         | 3                           | 4    |       |  |
| 1140   |     |          | RED ECC   | 5392450   | 3                            | 12   | REDUCER ECC SCH STD LTCS A420 WPL6            | 4                           | 14   |       |  |
| 1150   |     |          | RED ECC   | 5392291   | 8                            | 14   | REDUCER ECC SCH 20/STD LTCS A420 WPL6         | 16                          | 16   |       |  |
| 1160   |     |          | RED ECC   | 5392298   | 10                           | 14   | REDUCER ECC SCH 20/STD LTCS A420 WPL6-W       | 18                          | 20   |       |  |
| 1170   |     |          | RED ECC   | 5392288   | 16                           | 20   | REDUCER ECC SCH 20 LTCS A420 WPL6-W           | 18                          | 24   |       |  |
| 1180   |     |          | RED ECC   | 5392508   | 18                           | 24   | REDUCER ECC SCH STD/20 LTCS A420 WPL6-W       | 26                          | 36   |       |  |
| 1190   |     |          | RED ECC   | 5392509   | 26                           | 36   | REDUCER ECC SCH STD LTCS A420 WPL6-W          | 28                          | 38   |       |  |
| 1200   |     |          | RED ECC   | 5392510   | 30                           | 38   | REDUCER ECC SCH XS/STD LTCS A420 WPL6-W       | 40                          | 48   |       |  |
| 1210   |     |          | RED ECC   | 5392511   | 40                           | 46   | REDUCER ECC SCH XS LTCS A420 WPL6-W           | 42                          | 48   |       |  |
| 1220   |     |          |           |           |                              |      |                                               |                             |      |       |  |
| 1220   | U04 |          |           | 5544094   | 0.5                          | 0.5  | SWAGE CONC SCH XS/160 LTCS A420 WPL6 PLE/TSE  | 0.75                        | 0.75 |       |  |

|  | <b>TENGIZCHEVROI TCO, KAZAKHSTAN</b> | CONSTRUCTION                                                                                                                                     | CORR.ALLOW.         | SERVICE                                                                                                                 | MATERIAL                                                            | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|                                                                                    |                                      | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 2<br>of 4 | <b>150H21</b>     |

| SEQ NO | REV | NOTE NO. | ITEM NAME   | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                               | REFERENCE / LARGE END SIZES |    | NOTES |  |  |  |  |
|--------|-----|----------|-------------|-----------|------------------------------|------|-----------------------------------------------------------|-----------------------------|----|-------|--|--|--|--|
| 1230   | U04 |          | RED CONC    | 5544094   | 0.75                         | 0.75 | SWAGE CONC SCH XS/160 LTCS A420 WPL6 PLE/TSE              | 1                           | 1  |       |  |  |  |  |
| 1240   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1240   |     |          |             |           |                              |      | ***** OLETS *****                                         |                             |    |       |  |  |  |  |
| 1250   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1250   | U04 |          | OLETS       | 5504001   | 0.75                         | 1.5  | NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1              | 2                           | 2  |       |  |  |  |  |
| 1260   | U04 |          | OLETS       | 5504001   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1              | 40                          | 48 |       |  |  |  |  |
| 1270   | U04 |          | OLETS       | 5504002   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 3                           | 14 |       |  |  |  |  |
| 1280   | U04 |          | OLETS       | 5504002   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 26                          | 38 |       |  |  |  |  |
| 1290   | U04 |          | OLETS       | 5504016   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH 20/ XS LTCS A350 LF2 CL.1          | 16                          | 24 |       |  |  |  |  |
| 1300   | U04 |          | OLETS       | 5504020   | 0.75                         | 1.5  | NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1              | 1.5                         | 2  |       |  |  |  |  |
| 1310   | U04 |          | OLETS       | 5504021   | 0.75                         | 1    | NIPOFLANGE 300# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 8                           | 14 |       |  |  |  |  |
| 1320   | U04 |          | OLETS       | 5504021   | 0.5                          | 0.5  | NIPOFLANGE 300# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 30                          | 30 |       |  |  |  |  |
| 1330   | U04 |          | OLETS       | 5504021   | 0.75                         | 1.5  | NIPOFLANGE 300# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 3                           | 6  |       |  |  |  |  |
| 1340   | U04 |          | OLETS       | 5504021   | 0.75                         | 0.75 | NIPOFLANGE 300# RF SCH STD/ XS LTCS A350 LF2 CL.1         | 36                          | 38 |       |  |  |  |  |
| 1350   | U04 |          | OLETS       | 5504017   | 0.75                         | 0.75 | NIPOFLANGE 300# RF SCH 20/ XS LTCS A350 LF2 CL.1          | 16                          | 24 |       |  |  |  |  |
| 1360   | U04 | UJ2      | OLETS       | 5504025   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 4                           | 14 |       |  |  |  |  |
| 1370   | U04 | UJ2      | OLETS       | 5504025   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 26                          | 38 |       |  |  |  |  |
| 1380   | U04 | UJ2      | OLETS       | 5504019   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1          | 16                          | 24 |       |  |  |  |  |
| 1390   | U04 | UJ2      | OLETS       | 5504023   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1          | 40                          | 48 |       |  |  |  |  |
| 1400   | U04 | UJ3      | OLETS       | 5504325   | 3                            | 3    | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 8                           | 8  |       |  |  |  |  |
| 1410   | U04 | UJ3      | OLETS       | 5504325   | 3                            | 3    | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 30                          | 36 |       |  |  |  |  |
| 1420   | U04 | UJ4      |             | 5504326   | 2                            | 2    | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 6                           | 6  |       |  |  |  |  |
| 1430   | U04 | FF5      | OLETS       | 5504500   | 0.75                         | 1.5  | NIPOFLANGE FLAT BOTTOM 150# RF SCH XS LTCS A350 LF2 CL.1  | 0                           | 0  |       |  |  |  |  |
| 1440   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1440   | U04 |          | OLETS       | 5505335   | 0.75                         | 1    | WELDOLET SCH XS LTCS A350 LF2 CL.1                        | 2                           | 2  |       |  |  |  |  |
| 1450   | U04 |          | OLETS       | 5505179   | 0.75                         | 2    | WELDOLET SCH STD/ XS LTCS A350 LF2 CL.1                   | 3                           | 14 |       |  |  |  |  |
| 1460   | U04 |          | OLETS       | 5505179   | 0.75                         | 2    | WELDOLET SCH STD/ XS LTCS A350 LF2 CL.1                   | 26                          | 38 |       |  |  |  |  |
| 1470   | U04 |          | OLETS       | 5505572   | 0.75                         | 2    | WELDOLET SCH 20/ XS LTCS A350 LF2 CL.1                    | 16                          | 24 |       |  |  |  |  |
| 1480   | U04 |          | OLETS       | 5505333   | 0.75                         | 2    | WELDOLET SCH XS LTCS A350 LF2 CL.1                        | 40                          | 48 |       |  |  |  |  |
| 1490   | U04 |          | OLETS       | 5505164   | 3                            | 4    | WELDOLET SCH STD LTCS A350 LF2 CL.1                       | 8                           | 14 |       |  |  |  |  |
| 1500   | U04 |          | OLETS       | 5505164   | 3                            | 12   | WELDOLET SCH STD LTCS A350 LF2 CL.1                       | 26                          | 38 |       |  |  |  |  |
| 1510   | U04 |          | OLETS       | 5505571   | 3                            | 8    | WELDOLET SCH 20/ STD LTCS A350 LF2 CL.1                   | 16                          | 24 |       |  |  |  |  |
| 1520   | U04 |          | OLETS       | 5505334   | 3                            | 12   | WELDOLET SCH XS/ STD LTCS A350 LF2 CL.1                   | 40                          | 48 |       |  |  |  |  |
| 1530   | U04 |          | OLETS       | 5504374   | 0.75                         | 2    | WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1            | 0                           | 0  |       |  |  |  |  |
| 1540   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1540   | U04 | UJ2      | ELBOFLG     | 5516625   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 4                           | 14 |       |  |  |  |  |
| 1550   | U04 | UJ2      | ELBOFLG     | 5516625   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1         | 26                          | 38 |       |  |  |  |  |
| 1560   | U04 | UJ2      | ELBOFLG     | 5516619   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1          | 16                          | 24 |       |  |  |  |  |
| 1570   | U04 | UJ2      | ELBOFLG     | 5516623   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1          | 40                          | 48 |       |  |  |  |  |
| 1580   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1580   | U04 | 821      | OLETS       | 5509731   | 3                            | 3    | SWEEPOLET 19.05 MM/ STD LTCS A350 LF2 CL.1                | 24                          | 24 |       |  |  |  |  |
| 1590   | U04 | 821      | OLETS       | 5509721   | 3                            | 3    | SWEEPOLET SCH 20/ STD LTCS A350 LF2 CL.1                  | 16                          | 16 |       |  |  |  |  |
| 1600   | U04 | 821      | OLETS       | 5509720   | 1.5                          | 1.5  | SWEEPOLET SCH 20/ XS LTCS A350 LF2 CL.1                   | 16                          | 16 |       |  |  |  |  |
| 1610   | U04 | 821      | OLETS       | 5509724   | 1.5                          | 1.5  | SWEEPOLET STD/ XS LTCS A350 LF2 CL.1                      | 30                          | 30 |       |  |  |  |  |
| 1620   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1620   |     | FS1      |             |           |                              |      | ***** FLANGES *****                                       |                             |    |       |  |  |  |  |
| 1630   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1630   | U04 | FA8      | FLANGE      | 5599692   | 0.5                          | 0.75 | FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1       | 0.75                        | 2  |       |  |  |  |  |
| 1640   | U04 | FA8      | TEE RED     | 5599703   | 0.5                          | 0.5  | FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1       | 0.75                        | 2  |       |  |  |  |  |
| 1650   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1650   |     |          | FLG ADAPTER | 5851200   | 0.5                          | 2    | FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE                | 0                           | 0  |       |  |  |  |  |
| 1660   | U04 |          | FLG ADAPTER | 5851201   | 0.75                         | 0.75 | FLANGE ADAPTER 150# RF 316 SS X 18 MM TUBE                | 0                           | 0  |       |  |  |  |  |
| 1670   |     |          | FLANGE      | 5851209   | 1                            | 1    | FLANGE ADAPTER 150# RF 316 SS X 25 MM TUBE                | 0                           | 0  |       |  |  |  |  |
| 1680   | U04 |          | FLG ADAPTER | 5851203   | 0.5                          | 2    | FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE                | 0                           | 0  |       |  |  |  |  |
| 1690   |     |          |             |           |                              |      |                                                           |                             |    |       |  |  |  |  |
| 1690   | U04 |          | FLANGE      | 5556216   | 5                            | 5    | FLG WN 150# FF LTCS A350 LF2 CL.1 SCH STD                 | 0                           | 0  |       |  |  |  |  |
| 1700   | U04 |          | FLANGE      | 5556264   | 0.5                          | 2.5  | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS                  | 0                           | 0  |       |  |  |  |  |
| 1710   | U04 |          | FLANGE      | 5556261   | 3                            | 14   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD                 | 0                           | 0  |       |  |  |  |  |
| 1720   | U04 |          | FLANGE      | 5556265   | 16                           | 24   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20                  | 0                           | 0  |       |  |  |  |  |
| 1730   | U04 |          | FLANGE      | 5556262   | 26                           | 38   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD B16.47 SERIES A | 0                           | 0  |       |  |  |  |  |
| 1740   | U04 |          | FLANGE      | 5556266   | 40                           | 48   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A  | 0                           | 0  |       |  |  |  |  |
| 1750   | U04 |          | FLANGE      | 5556476   | 0.5                          | 2    | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS                  | 0                           | 0  |       |  |  |  |  |
| 1760   | U04 |          | FLANGE      | 5556473   | 3                            | 14   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD                 | 0                           | 0  |       |  |  |  |  |
| 1770   | U04 |          | FLANGE      | 5556474   | 16                           | 24   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20                  | 0                           | 0  |       |  |  |  |  |
| 1780   | U04 |          | FLANGE      | 5556475   | 26                           | 38   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD B16.47 SERIES A | 0                           | 0  |       |  |  |  |  |
| 1790   | U04 |          | FLANGE      | 5556477   | 40                           | 48   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A  | 0                           | 0  |       |  |  |  |  |
| 1800   | U04 |          | FLANGE      | 5557036   | 0.75                         | 2    | FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS                  | 0                           | 0  |       |  |  |  |  |
| 1810   | U04 |          | FLANGE      | 5557033   | 3                            | 12   | FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD                 | 0                           | 0  |       |  |  |  |  |
| 1820   | U04 |          | FLANGE      | 5555935   | 3                            | 6    | FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD                 | 0                           | 0  |       |  |  |  |  |
| 1830   | U04 |          | FLANGE      | 5557393   | 0.75                         | 2    | FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS                 | 0                           | 0  |       |  |  |  |  |
| 1840   | U04 | 817      | FLANGE      | 5557704   | 0.75                         | 2    | FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XS                | 0                           | 0  |       |  |  |  |  |
| 1850   | U04 |          | BLIND       | 5615566   | 0.5                          | 24   | FLG BLIND 150# RF LTCS A350 LF2 CL.1                      | 0                           | 0  |       |  |  |  |  |
| 1860   | U04 |          | BLIND       | 5615570   | 26                           | 48   | FLG BLIND 150# RF LTCS A350 LF2 CL.1 B16.47 SERIES A      | 0                           | 0  |       |  |  |  |  |

|  | <b>TENGIZCHEVROIL<br/>TCO<br/>TENGIZ,<br/>KAZAKHSTAN</b> | CONSTRUCTION                                                                                                                                     | CORR.ALLOW.         | SERVICE                                                                                                                 | MATERIAL                                                            | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|                                                                                    |                                                          | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 3<br>of 4 | <b>150H21</b>     |

| SEQ NO | REV | NOTE NO. | ITEM NAME    | ITEM CODE | SIZE RANGE / SMALL END SIZES |     | DESCRIPTION                                                         | REFERENCE / LARGE END SIZES |   | NOTES |
|--------|-----|----------|--------------|-----------|------------------------------|-----|---------------------------------------------------------------------|-----------------------------|---|-------|
| 1870   | U04 |          | BLIND        | 5615576   | 0.5                          | 24  | FLG BLIND 300# RF LTCS A350 LF2 CL.1                                | 0                           | 0 |       |
| 1880   | U04 |          | BLIND        | 5615579   | 26                           | 48  | FLG BLIND 300# RF LTCS A350 LF2 CL.1 B16.47 SERIES A                | 0                           | 0 |       |
| 1890   | U04 |          | BLIND        | 5615598   | 1                            | 6   | FLG BLIND 600# RF LTCS A350 LF2 CL.1                                | 0                           | 0 |       |
| 1900   | U04 |          | BLIND        | 5615605   | 3                            | 6   | FLG BLIND 900# RF LTCS A350 LF2 CL.1                                | 0                           | 0 |       |
| 1910   | U04 |          | BLIND        | 5615616   | 0.75                         | 2   | FLG BLIND 1500# RF LTCS A350 LF2 CL.1                               | 0                           | 0 |       |
| 1920   | U04 | 817      | BLIND        | 5615615   | 2                            | 2   | FLG BLIND 2500# RTJ LTCS A350 LF2 CL.1                              | 0                           | 0 |       |
| 1930   |     |          |              |           |                              |     | *****ORIFICE FLANGES*****                                           |                             |   |       |
| 1930   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 1940   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 1940   | U04 |          | ORIFICE FLG  | 5623947   | 2                            | 2   | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS                       | 0                           | 0 |       |
| 1950   | U04 |          | ORIFICE FLG  | 5623944   | 3                            | 14  | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD                      | 0                           | 0 |       |
| 1960   | U04 |          | ORIFICE FLG  | 5623941   | 16                           | 24  | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 20                       | 0                           | 0 |       |
| 1970   | U04 |          | ORIFICE FLG  | 5624801   | 26                           | 38  | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD ASME B16.47 SERIES A | 0                           | 0 |       |
| 1980   | U04 |          | ORIFICE FLG  | 5624800   | 40                           | 48  | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS ASME B16.47 SERIES A  | 0                           | 0 |       |
| 1990   |     |          |              |           |                              |     | ***** PIPE NIPPLES                                                  |                             |   |       |
| 1990   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2000   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2000   | U04 |          | NIPPLE       | 5541197   | 0.75                         | 2   | NIPPLE SCH 160 LTCS A333 GR.6 BOE/TOE 3" LONG                       | 0                           | 0 |       |
| 2010   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2010   |     | FS1      |              |           |                              |     | ***** GASKETS *****                                                 |                             |   |       |
| 2020   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2020   |     |          | GASKET       | 5673043   | 0.5                          | 24  | GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR                          | 0                           | 0 |       |
| 2030   |     |          | GASKET       | 5674011   | 26                           | 48  | GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A         | 0                           | 0 |       |
| 2040   |     |          | GASKET       | 5673033   | 0.5                          | 24  | GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR                          | 0                           | 0 |       |
| 2050   | U04 | GA6      | GASKET       | 5673017   | 2                            | 3   | GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5                   | 0                           | 0 |       |
| 2060   |     |          | GASKET       | 5674032   | 26                           | 48  | GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A         | 0                           | 0 |       |
| 2070   | U04 |          | GASKET       | 5673103   | 0.75                         | 20  | GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR                          | 0                           | 0 |       |
| 2080   | U04 |          | GASKET       | 5673117   | 3                            | 6   | GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR                          | 0                           | 0 |       |
| 2090   |     |          | GASKET       | 5672251   | 0.75                         | 2   | GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR                         | 0                           | 0 |       |
| 2100   |     |          | GASKET       | 5669711   | 5                            | 5   | GASKET 150# FF NON-ASB PART AGT 1/16"THK                            | 0                           | 0 |       |
| 2110   | U04 | UG1      | GASKET       | 5664201   | 2                            | 24  | GASKET 150# RF FLAT RING PTFE                                       | 0                           | 0 |       |
| 2120   |     |          | GASKET       | 5672271   | 2                            | 2   | GASKET 2500# 316SS S PIRAL WOUND GRAPHITE IR                        | 0                           | 0 |       |
| 2130   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2130   |     | B1       |              |           |                              |     | ***** BOLTING *****                                                 |                             |   |       |
| 2140   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2140   | U04 |          | BOLTING      | 5675770   | 0.5                          | 99  | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS                   | 0                           | 0 |       |
| 2150   | U04 | UG2      | BOLTING      | 5675768   | 0.5                          | 99  | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS      | 0                           | 0 |       |
| 2160   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2160   |     | SB6      | BOLTING      | 5675675   | 12                           | 48  | JACK SCREW HEX HEAD A320 GR L7M                                     | 0                           | 0 |       |
| 2170   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2170   |     | FS1      |              |           |                              |     | **** BLINDS & SPACERS ****                                          |                             |   |       |
| 2180   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2180   |     |          | SPEC BLIND   | 5749580   | 0.75                         | 14  | SPECTACLE BLIND 150# RF LTCS A 516 GR.70                            | 0                           | 0 |       |
| 2190   | U04 |          | SPEC BLIND   | 5749581   | 0.75                         | 10  | SPECTACLE BLIND 300# RF LTCS A516 GR.70                             | 0                           | 0 |       |
| 2200   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2200   |     |          | PADDLE       | 5750023   | 16                           | 24  | PADDLE BLANK 150# RF LTCS A516 GR.70                                | 0                           | 0 |       |
| 2210   |     |          | PADDLE       | 5750100   | 12                           | 24  | PADDLE BLANK 300# RF LTCS A516 GR.70                                | 0                           | 0 |       |
| 2220   |     |          | PADDLE       | 5750240   | 16                           | 24  | PADDLE SPACER 150# RF LTCS A516 GR.70                               | 0                           | 0 |       |
| 2230   |     |          | PADDLE       | 5750200   | 12                           | 24  | PADDLE SPACER 300# RF LTCS A516 GR.70                               | 0                           | 0 |       |
| 2240   |     |          | PADDLE       | 5750026   | 26                           | 48  | PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A                 | 0                           | 0 |       |
| 2250   |     |          | PADDLE       | 5750028   | 26                           | 48  | PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A                 | 0                           | 0 |       |
| 2260   |     |          | PADDLE       | 5750196   | 26                           | 48  | PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A               | 0                           | 0 |       |
| 2270   |     |          | PADDLE       | 5750198   | 26                           | 48  | PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A                | 0                           | 0 |       |
| 2280   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2290   |     |          |              |           |                              |     | ***** COUPLINGS, UNIONS & PLUGS *****                               |                             |   |       |
| 2300   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2300   | U04 | ND9      | Coupling     | 5478705   | 1.5                          | 1.5 | COUPLING 3000# SW LTCS A 350 LF2                                    | 0                           | 0 |       |
| 2310   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2310   | U04 |          | FITTING SCRD | 5491303   | 0.5                          | 0.5 | PLUG ROUND HEAD SCRD LTCS A350 LF2 CL.1                             | 0                           | 0 |       |
| 2320   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2320   |     |          |              |           |                              |     | *****MISCELLANEOUS*****                                             |                             |   |       |
| 2330   |     |          |              |           |                              |     |                                                                     |                             |   |       |
| 2330   | U04 | 823      | FLG GUARD    | 5558881   | 0.5                          | 10  | SAFETY SHIELD PTFE SUITABLE FOR 150# FLANGE                         | 0                           | 0 |       |

|                                                                                    |                                                          |                                                                                                                                                  |                     |                                                                                                                         |                                                                     |             |            |            |                |                          |
|------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------|------------|------------|----------------|--------------------------|
|  | <b>TENGIZCHEVROIL<br/>TCO<br/>TENGIZ,<br/>KAZAKHSTAN</b> | <b>CONSTRUCTION</b>                                                                                                                              | <b>CORR.ALLOW.</b>  | <b>SERVICE</b>                                                                                                          | <b>MATERIAL</b>                                                     | <b>DATE</b> | <b>BY</b>  | <b>REV</b> | <b>SHEET</b>   | <b>PIPING LINE CLASS</b> |
|                                                                                    |                                                          | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW: 1.5 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024  | BY<br>OZAC | U04        | Page 4<br>of 4 | <b>150H21</b>            |